Paul Parisot

Software Engineer - Core C++ & Performance Computing

Github: paulogarithm — paul.parisot@epitech.eu - +33 6 42 36 01 57

About

Performance-driven software engineer specialized in C/C++ systems programming and low-level optimization.
Strong foundations in applied physics/mathematics for 3D environments, including matrix transformations and vectorization.
Experienced in industrial embedded environments, writing clean, modular, and highly efficient code under Linux.

Skills

Technical Expertise: C, C++ (Modern standards up to C++20/23), Assembly, Go, Linux Systems
Math & Optimization: 3D Math (4x4 Matrices, Coordinate Systems), SIMD Vectorization, Collision Algorithms
Protocols & Tooling: TCP/IP Sockets, Git Toolchain, DevOps (Docker, Kubernetes), CI/CD pipelines
Methodology: Agile/Scrum workflow, Jira task tracking, Strict Code Review compliance

Professional Experience

Faurecia / Forvia - Software Engineer (Embedded C++ / Agile)

- Developed low-level C++ network modules for embedded communications, ensuring reliable data exchange.
- Engineered a real-time navigation and path-generation system showcased globally at **CES 2024**.
- Worked within a structured Agile/Scrum environment, utilizing **Jira** for tracking requirements and deliveries.
- Formulated clean, robust architectures praised by management for their speed and edge-case coverage.

Gno.land - Junior Software Engineer (Open-Source Core)

- Developed 6+ core packages in an open-source runtime environment, focusing on memory efficiency and safety.
- Fixed **6 critical runtime crash bugs**, contributing over **10k lines of code** to core packages.
- Wrote comprehensive regression and integration tests under automated, zero-tolerance CI/CD pipelines.

Epitech / IONIS - Technical Assistant & Instructor

- Mentored over **400 students** in low-level concepts: C, C++, Assembly, and algorithm design.
- Enforced strict industrial coding standards, code quality metrics, and algorithmic complexity constraints.
- Drafted technical documentation, tutorials, and advanced evaluation content for junior student cohorts.

Technical Projects & Applied Mathematics

rodox – Modular Systems & Render Engine under Linux

Designed a low-level dynamic engine architecture in C/C++ natively on Linux using an ECS architecture.
Implemented 3D mathematical layers using **4x4 transformation matrices (CFrames)** and linear algebra.
Optimized data layouts and compute bottlenecks using hardware-accelerated **SIMD vectorization** pipelines.
Skills: C++, Linux OS, 3D Linear Algebra, SIMD Optimization, Memory Management, Dynamic Linking

Raytracer – 3D Mathematical Renderer from Scratch

Developed a software-based light transport engine in C++ focused on mathematical modeling.
Coded vector calculus operations, matrix inversions, and high-frequency **ray-object collision algorithms**.
Skills: C++, Geometry Algorithms, Collision Detection, Linear Algebra, Object-Oriented Design

Theoretical Foundations & Core Research

- **Advanced Computer Science**: Rigorous study of theoretical frameworks (automata theory, advanced complexity, and formal languages).
- **Low-Level Systems**: Engineered a runtime assembly-patching engine in pure C; dynamically injected instance context addresses into compiled binary stubs executed on-the-fly via allocated `mmap` pages. (open source, on my github)
- **Project Management**: Acquired formal methodologies in team alignment, scope definition, and strategic risk assessment.

Education

Epitech, Paris 2022 – 2027
Master's Degree in Information Technology (Level-7 RNCP Certification)
Specialization: Systems Engineering and Advanced Software Architecture

McGill University, Montréal 2025 – 2026
Certificate in Management
Complementary coursework in foundational physics/engineering contexts and administration.